

Python Machine Learning (CSCI 425) Student Showcase (Final Project) Rubric

Project Title:

Instructions:

1. For each requirement, use 0.0 - 5.0 scale in the Score column (0 - Fail, 1 - Needs improvement, 2 - Poor, 3 - Fair, 4 - Good, 5 is Excellent)

Criteria	Score (1-5)
1. Problem Definition & Motivation	
a. Clearly defined problem statement.	5
b. Understanding of the domain and context	4
2. Data Acquisition & Preprocessing	
a. Data cleaning and handling missing values	5
b. Feature engineering and selection	4
c. Data normalization and scaling	5
3. Model Selection, Evaluation & Justification	
a. Selection of appropriate ML algorithms	4
b. Model training and tuning	5
c. Evaluation metrics and performance analysis	5
4. Creativity and Innovation	
a. Novelty and originality of approach	3
b. Exploration of advanced techniques (Deep Learning, e.g.)	3
5. Presentation	
a. Quality of visualizations and insights	5
b. Ability to communicate results effectively	4
Final Score	52/60

General Feedback:

- Overall, I felt that our group did a strong job with the core parts of the project, especially cleaning the data, evaluating the dataset, comparing model performance, and choosing a model that matched the cybersecurity problem. I think our project was successful because we focused on practical effectiveness rather than only trying to make the model more complex. If we had more time, I would have liked to explore additional machine learning algorithms, try more advanced techniques, and further develop the web application. Overall, I felt this was a well-established project that focused on the most important aspects of building an effective intrusion detection system.